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ORIGINAL PAPER

Study on Tribological Properties of Gear Material PA66 Reinforced by Glass Fibers

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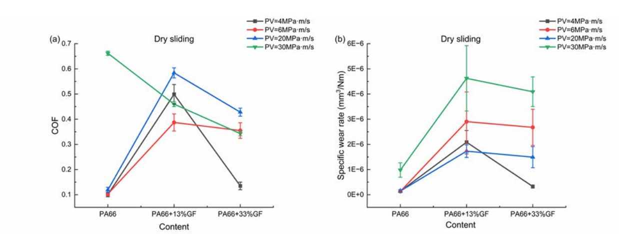
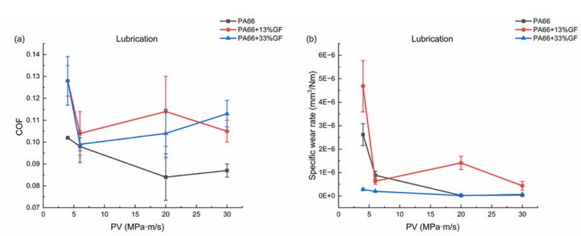
Abstract

Polyamide 66 (PA66) is one of the commonly used polymer gear materials. This paper focuses on the tribological properties of glass fiber reinforced PA66 composites in self-mated contact using a pin-on-disc tribometer. The effects of glass fiber content, PV (the product of the contact pressure and sliding speed), and lubrication on the tribological properties of the specimens are also investigated. The results show that the glass fiber reinforced PA66 exhibit higher coefficients of friction and specific wear rates than PA66 under dry sliding conditions. This is probably due to the peeled glass fibers during the sliding process acting as abrasive particles which have an aggressive effect on the surface. Under grease lubricated conditions, PA66 + 33% GF has the lowest coefficient of friction and specific wear rate due to its higher strength. Under dry sliding conditions, all specimens show the highest friction coefficient and specific wear rate at 30 MPa·m/s with the change of PV value. Under grease lubricated conditions, all specimens show the highest friction coefficient and specific wear rate at 4 MPa·m/s with the change of PV value. The addition of grease improves friction and wear of PA66 composites under most of the experimental conditions. However, the specific wear rates of PA66 and PA66 + 13% GF under grease lubrication are higher than those under dry sliding conditions at low PV values. This may be due to the fact that greases can reduce the surface mechanical strength of PA66 and PA66 + 13% GF.

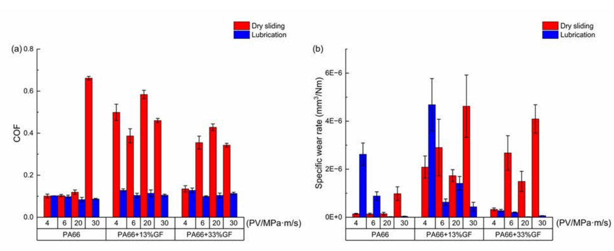
Graphical abstract



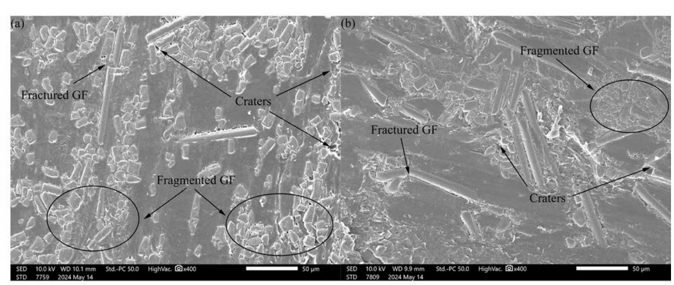
Effect of glass fiber content on tribological properties of PA66 composites



Effect of PV values on tribological properties of PA66 composites



Effect of grease lubrication on tribological properties of PA66 composites



SEM images of worn surfaces of PA66 composites under dry sliding conditions: (a) PA66 + 13% GF, (b) PA66 + 33% GF (Contact pressure: 60 MPa, sliding speed: 0.5 m/s)

Keywords PA66 · Glass fiber · Friction · Wear · PV · Grease lubrication · Wear mechanism

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1 Introduction

PA66 is widely used for polymer gears due to its self-lubricating, light weight, wear resistance, and high temperature resistance [1, 2]. To improving the ability of polymer gears under more severe conditions could widen their application, such as downsize and decrease the weight of gears, it is necessary to improve the mechanical properties and wear resistance of PA66. Usually, adding reinforcing agents such as glass fiber, carbon fiber, MoS_2 can improve the mechanical or tribological properties of PA66. Glass fiber typically enhances the strength of polymer materials [3, 4] but its effect on the tribological properties of PA66 has not been clearly established. Furthermore, grease is usually used to reduce friction, wear, and the heat generated in gear transmission system [5, 6]. Therefore, an in-depth investigation of the effects of glass fiber reinforcement and grease lubrication on the tribological properties of PA66 is essential to advance the scientific understanding and practical application of this polymeric material.

Glass fibers can improve the strength of polymer matrix, and many scholars have investigated the effects of glass fiber content, orientation, and sliding conditions on tribological properties of polyamide composites. Chen et al. [7, 8] reported that glass fibers improved the mechanical properties of the blends but did not significantly improve the wear resistance of the blends. Whereas some scholars [9–13] reported that glass fibers effectively reduced the coefficient of friction and wear rate of polyamide composites. In addition, some studies [14, 15] have investigated the effects of fiber diameter, molecular weight of the polymer, and different average molar masses on the tribological properties of PA66 composites. Kunishima et al. [14] reported that the interfacial adhesion of the glass fibers to the PA66 matrix improves the wear and creep resistance of the composites and reduce the aggressive effect of glass fibers on the steel counterpart. ShinS et al. [15] also reported that the weight average molar mass of composites can change its shear strength and adhesion to improve the tribological performance. The mechanical properties and tribological behavior of PA66 are affected by water absorption which changes its molecular structure, thereby reducing strength and dimensional stability. Kim et al. [16] investigated the effect of short glass fiber reinforcement on the friction and wear properties of PA66 under humid conditions. They reported that the short glass fiber reinforcement could produce a contact plateau to increase the density of fibers and thus effectively improve the friction and wear properties of PA66. Daniel et al. [17] reported that glass fibers could improve the resistance to water absorption and tribological properties of PA66 in sea water.

To elucidate the effects of contact pressure and sliding speed on tribological properties of PA66 composites. Myshkin et al. [18] reported that loads and velocities affect the mechanical properties of polymers. The existence of a critical load or critical velocity causes the material to transition between plastic and elastic contact. Mihai et al. [19] reported that an increase in load leads to softening of the surface layer and a decrease in the shear strength of PA66 which leads to a decrease in the coefficient of friction. Yu et al. [20] reported that the coefficient of friction decreases with increasing load, while the wear increases with increasing load, they also found that the transfer film significantly enhances the tribological properties of PA66 composites. While some studies [10, 21, 22] have reported that the coefficient of friction of PA composites increases with increasing load. Some studies [23, 24] reported that lower sliding speeds can achieve lower wear. Kalin et al. [25] reported that plastic deformations occurred on PA6 surface at high speeds due to the poor thermal conductivity. Unal et al. [22] reported similar results. Ana et al. [26] reported that an increase in sliding velocity makes the glass fibers detach more easily from the polymer matrix, leading to an increase in the fiber content of the sliding surfaces. This leads to a decrease in the coefficient of friction. Neis et al. [27] reported that if the material exceeds its PV limit value, it could lead to severe wear.

Most of the aforementioned studies focused on investigating the tribological properties of PA66 composites under dry sliding conditions. In contrast, the tribological properties of PA66 composites under lubricating conditions have been less extensively studied. Liu et al. [21] reported that the coefficients of friction and wear rates of PA composites under oil-in-water lubricated sliding conditions were much lower than those under dry sliding conditions. Daniel et al. [17] reported that the insoluble salts can act as lubricating film to improve the tribological properties of PA66 composites in different seawater solutions. Sanes et al. [28] reported that ionic liquid lubricants have the effect of improving the tribological properties of PA6. Kunishima et al. [12–14] investigated the tribological properties of glass fibers reinforced PA66 and also found similar results under grease lubricated conditions. While Jia et al. [29] investigated the tribological properties of PA66-PA66 under dry sliding and oil lubricated conditions, and found that the antiwear properties of PA66 were reduced by the oil lubricant.

Previous studies have predominantly investigated the tribological properties of PA66 composites in contact with a metallic material under dry conditions, with a focus on flat–flat contact. There are few studies regarding the tribological properties of glass fiber reinforced PA66 in self-mated contact under Hertzian contact and grease lubricated condition. It is well known that gear meshing involves both sliding and rolling but friction and wear is mainly caused

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Table 1 The properties of materials

Specification	Value		
Material	PA66	PA66 + 13% GF	PA66 + 33% GF
Density, g/cm^3	1.14	1.23	1.39
Melting temperature, °C	262	262	262
Glass transition temperature, °C	60	80	80
Tensile modulus, MPa	3100	5500	11,000
Bending modulus, MPa	2800	4900	9500

Table 2 The properties of MOLYKOTE® EM-50L

Grease name	MOLYKOTE® EM-50L
Base oil	PAO
Thickener	Lithium
Base oil viscosity at 40 ℃(cSt)	1050
NLGI level	1
Color	translucent white
High temperature (℃)	150
Low temperature (℃)	−40

by the sliding part. Li et al. [30] compared the trend of gear mesh torque from FZG gear experiment and friction coefficient from pin-on-disc experiment and showed that pin-on-disc can simulate the sliding part of gear mesh. This study simulates the Hertzian contact and sliding part of gear mesh under both dry sliding and grease lubricated conditions using a pin-on-disk tribometer. The effects of glass fiber content, PV, and lubrication on the tribological properties of PA66 composites in self-mated contact are studied. The damage mechanism of polymer worn surfaces was also studied.

2 Experimental

2.1 Materials

The polymer materials used in this study were PA66 (DuPont, USA, PA66 101L), PA66 + 13% GF (DuPont, USA, 70G13HS1L), PA66 + 33% GF (DuPont, USA, 70G33HS1L), the glass fibers were 9–13 μ m in diameter and the length is about 1 mm, and the properties of the PA66 and its composites are shown in Table 1. The specimens were produced using a twin-screw extruder (TDS-35C) followed by an injection molding machine (JDW J50ADS). The extruder had different temperature zones (240–270℃) and screw speed was set to 80 rpm. During the injection molding process, the barrel temperature was maintained at 290℃, and the mold temperature was maintained at 100℃. The injection pressure was 120 MPa, and the holding pressure was 60 MPa. After the injection molding process, the sample is demolded and cooled to ambient temperature. The grease used in the study was (MOLYKOTE® EM-50L), and the properties of the EM-50L are shown in Table 2. The hardness of the three materials before and after grease immersion was tested using a nanoindentation meter (Bruker TI980) and the results are shown Table 1. The grease is suitable for the working scenarios of PA66 composite gears.

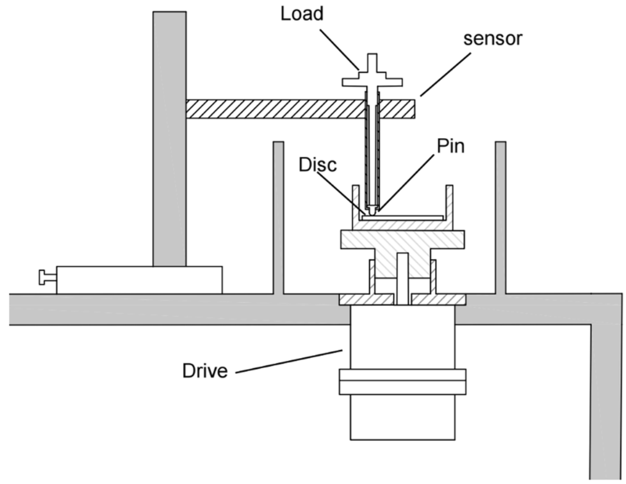

Fig. 1 Schematic diagram of the pin-on-disc tribometer

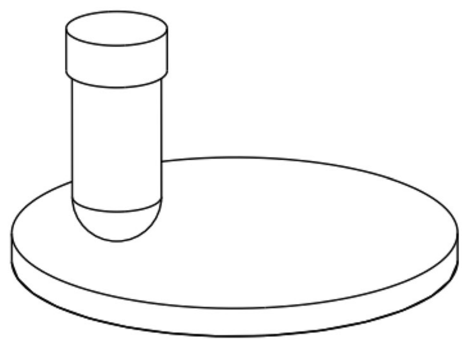

Fig. 2 Pins and discs used in the experiment

2.2 Experimental Details

The wear tests were performed using a pin-on-disc tribometer as shown in Fig. 1. The size of the pin is Ø 6 mm × 15 mm, and the size of the disc is Ø 30 mm × 2 mm, as shown in Fig. 2. The surface roughness of disc is Ra = 1.6 μm. The surface roughness of the pin samples made of PA66, PA66 + 13% GF, and PA66 + 33% GF are 0.22 μm, 3.6 μm, and 1.6 μm, respectively. The friction and wear tests were

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Table 3 Test conditions of friction and wear test

Parameters	value
Test temperature	Room temperature (25 ± 2 ℃)
Contact pressure	40 MPa, 60 MPa
Sliding velocity	0.1 m/s, 0.5 m/s
PV values	4 MPa·m/s (40 MPa-0.1m/s), 6 MPa·m/s (60 MPa-0.1m/s), 20 MPa·m/s (40 MPa-0.5m/s), 30 MPa·m/s (60 MPa-0.5m/s)
Grease	EM-50L

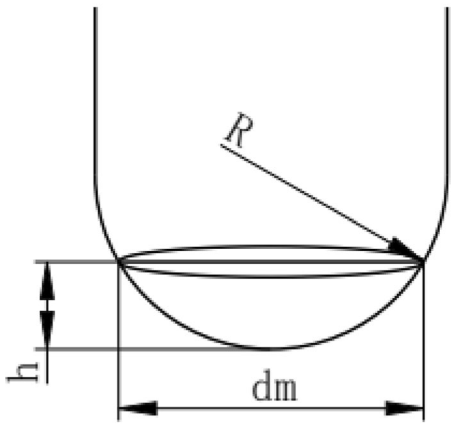

Fig. 3 The wear scar

conducted at room temperature under a normal load of 1.2 N or 3.9 N (equivalent to a maximum Hertzian contact pressure of 40 MPa or 60 MPa) and operated at a sliding speed of 0.1 m/s or 0.5 m/s (120 rpm or 600 rpm). According to the references [25, 31], the experimental time was set to 20 h to achieve steady state under polymer–polymer self-mated contact. Experiments were conducted under both dry sliding and grease lubricated conditions. The specific parameters of the experiment are shown in Table 3. All samples were dried at 70℃ for 24 h before the experiment. Each set of experimental conditions was repeated three times. The coefficient of friction was calculated as the measured friction force divided by the normal load applied to the pin. Since the loads in this paper are relatively small and not enough for creep to occur, the effect of creep on wear is ignored. The diameter of worn surface was measured using an optical microscope (Nikon/LV 150N) after experiment. The wear volume loss of the pin can be calculated using the following method. The main parameters for this calculation are shown in Fig. 3.

The wear volume loss V of the hemispherical pin is calculated by the following formula:

$$V = \pi h^2\left(R - \frac{h}{3}\right) \tag{1}$$

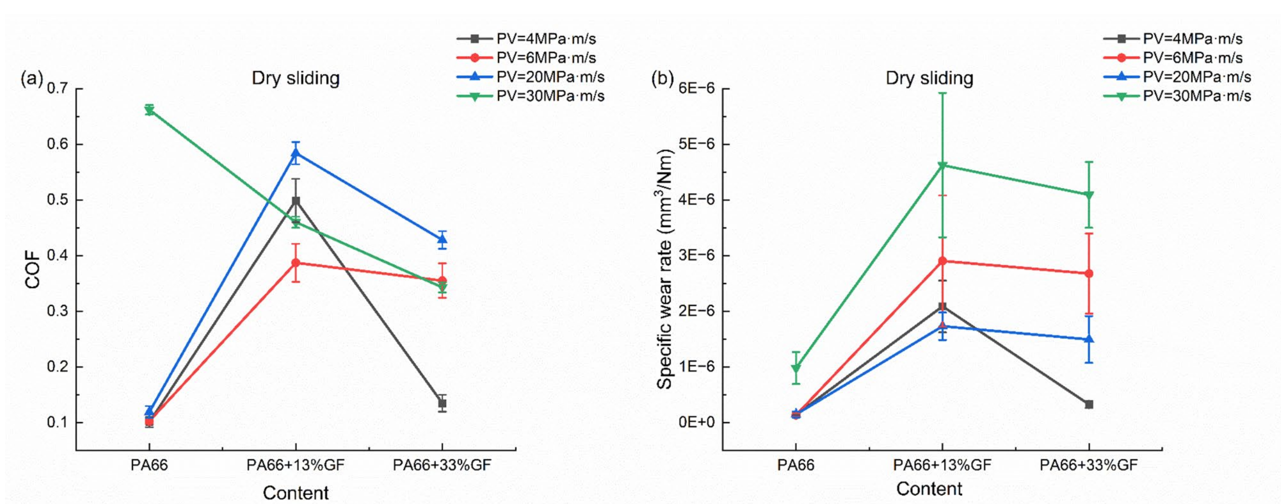

Fig. 4 The tribological properties of PA66 composites under dry sliding conditions: **a** Coefficient of friction; **b** Specific wear rate of pin specimen

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where, d_m is the diameter of the worn surface, R is the radius of the hemisphere, and h is the height of the wear volume loss. The calculation formula of h is as follows:

$$h = R - \sqrt{R^2 - \left(\frac{d_m}{2}\right)^2} \tag{2}$$

Calculate the specific wear rate of pin specimen according to Archard's wear model:

$$K = \frac{V}{NS} \tag{3}$$

where, S is the sliding distance, and N is the normal load.

The worn surface of samples was examined using a scanning electron microscope (SEM IT500). Prior to the examination, the pin specimens were sputter coated with Au for 60 s. All the characterization work mentioned above was completed at the Songshan Lake Materials Laboratory in Dongguan, China.

3 Results and Discussion

3.1 Effect of Glass *Fiber* Content on the Tribological Properties of PA66

Figure 4 shows the coefficient of friction and specific wear rate of specimens with different glass fiber contents under dry sliding conditions. The figure indicates that as the glass fiber content increases, both the specific wear rate and coefficient of friction of the samples exhibit a similar trend. The coefficients of friction and specific wear rates of the samples initially increase with the increase in glass fiber content, and then decrease. PA66 exhibits the lowest coefficient of friction and specific wear rate, while PA66 + 13% GF exhibits the highest. This means that glass fiber reinforced PA66 shows higher coefficients of friction and specific wear rates than PA66. According to the references [9–13], glass fibers can effectively improve the tribological properties of PA66. However, the positive effect of glass fibers on friction and wear was not observed in the polymer–polymer self-mated contact in this study. This may be attributed to the inherent differences between polymer–polymer contact combinations and polymer-metal contact combinations. Furthermore, fractured and fragmented glass fibers, as well as craters formed by glass fibers peeled off from the polymer matrix can be observed in Fig. 5. It can be inferred that the pulled-out glass fibers act as abrasive particles during the sliding process [14, 17, 32, 33], producing aggressive effects on the surface of disk. This causes the glass fiber of the disk to pull out, which may cause further and more serious damage to sliding surfaces and may explain the higher coefficient of friction and specific wear rate of glass fiber reinforced PA66. Moreover, PA66 + 33% GF exhibits lower coefficients of friction and specific wear rates than those of PA66 + 13% GF. A comparison of the SEM images of PA66 + 13% GF and PA66 + 33% GF shows that there are relatively fewer pulled-out glass fibers on the surface of PA66 + 33% GF. According to the references [32, 34, 35], the friction component resulting from adhesion is equal to the product of the real contact area and the polymer shear strength. Glass fiber reinforcement can improve the strength of the PA66 composites [7, 8], potentially leading to the higher coefficient of friction observed in glass fiber reinforced PA66. In addition, in PA66/GF-PA66/GF contacts, friction component resulting from adhesion include polymer–polymer,

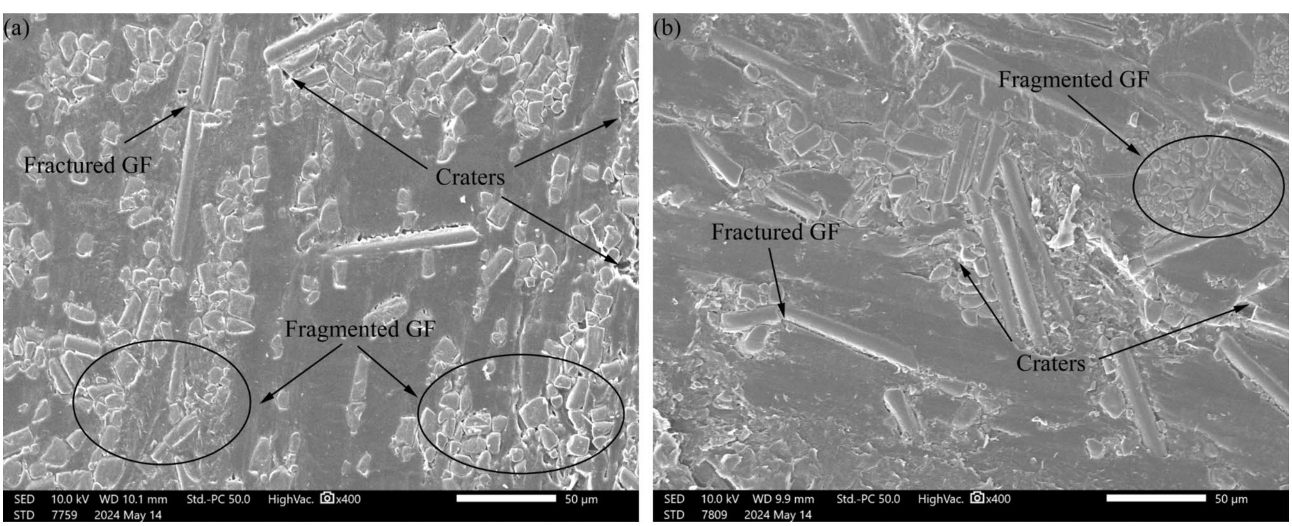

Fig. 5 SEM images of worn surfaces of PA66 composites pins under dry sliding conditions: **a** PA66 + 13% GF, **b** PA66 + 33% GF (Contact pressure: 60 MPa, sliding speed: 0.5 m/s)

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polymer-glass fiber, and glass fiber-glass fiber. This may also lead to an increase in the real contact area in the glass fiber reinforced PA66-glass fiber reinforced PA66 contact combinations, resulting in a higher coefficient of friction. Additionally, the hard glass fibers that are pulled out during the sliding process could induce the plowing effect on the sliding surface. The peeling glass fibers may cause damage to the matrix of the countersurface, resulting in transfer of the polymer matrix and higher wear. The interaction between the glass fibers may further increase friction and wear due to the fact that both surfaces involved in sliding with produce peeling glass fibers. Therefore, the glass fibers that peel off from the two mating surfaces may have an adhesion and plowing effect on the contact surface, which may also be a reason for the higher coefficient of friction of glass fiber reinforced PA66 under most experimental conditions. Figure 5 shows that PA66 + 13% GF produces more pulled-out glass fibers on the wear surface, which could cause a stronger plowing effect on the sliding surface, which is why PA66 + 13%GF exhibits a higher coefficient of friction and wear than PA66 + 33% GF. However, at a PV of 30 MPa·m/s, the coefficient of friction decreases with increasing glass fiber content, with PA66 exhibiting the highest coefficient of friction. As speed and contact pressure increase, the contact temperature typically rises due to frictional heating, which has two contrary effects on the coefficient of friction. On the one hand, increased temperature leads to specimen softening and a reduction in shear strength [35], thereby reducing the coefficient of friction. On the other hand, the elastic modulus of the polymers decreases at higher temperatures, increasing the real contact area and thus limiting the reduction in the coefficient of friction. In this study, the high friction coefficient at higher speed and contact pressure may be attributed to the dominant effect of the increased real contact area.

Figure 6 shows the coefficient of friction and specific wear rate of the three specimens with different glass fiber contents under grease lubricated conditions. Regarding the coefficient of friction of the specimens under grease lubricated conditions, it can be seen that the coefficients of friction of the glass fiber reinforced PA66 are higher than those of PA66. The results are similar to those under dry sliding conditions. This is because glass fiber reinforced PA66 exhibits a higher shear strength compared to PA66, resulting in a higher coefficient of friction. At PV values of 4 MPa·m/s and 20 MPa·m/s, PA66 + 33% GF shows the lowest specific wear rate, while the highest specific wear rate is observed for PA66 + 13% GF. At a PV of 6 MPa·m/s, PA66 + 33% GF has the lowest specific wear rate, and the specific wear rate of PA66 and PA66 + 13% GF is 4.4 times and 3.1 times higher than that of PA66 + 33% GF, respectively. This may be due to the fact that the grease leads to a reduction in the strength of all three specimens [15, 29, 36]. The softening effect of the grease has a greater weakening effect on PA66 and PA66 + 13% GF, resulting in lower wear for PA66 + 33% GF. At a PV of 6 MPa·m/s, PA66 showed the highest specific wear rate due to PA66 is weakened by grease more severely. At a PV of 30 MPa·m/s, PA66 has the lowest specific wear rate, and the specific wear rate of PA66 + 13% GF and PA66 + 33% GF is 11 times and 1.6 times higher than that of PA66, respectively. As shown in Fig. 7, the glass fibers in PA66 + 13% GF and PA66 + 33% GF began to peel off from the polymer matrix due to increased contact pressure at high PV values. As in the case of dry sliding, the glass fibers peeled off have an aggressive effect on the counterpart surface, resulting in a higher specific wear rates for glass fiber reinforced PA66 at a PV of 30 MPa·m/s [12–14].

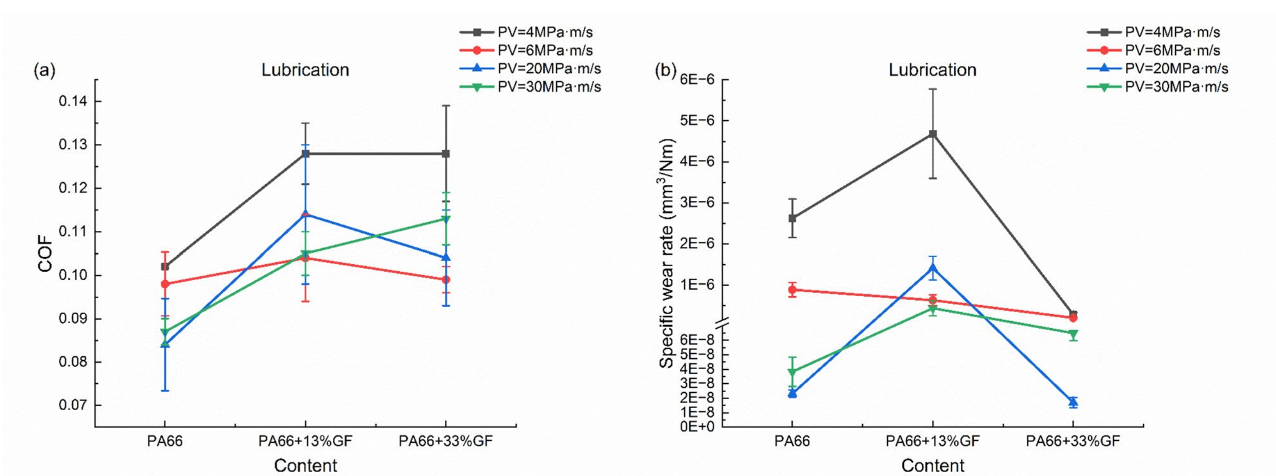

Fig. 6 The tribological properties of PA66 composites under grease lubricated conditions: **a** Coefficient of friction; **b** Specific wear rate of pin specimen

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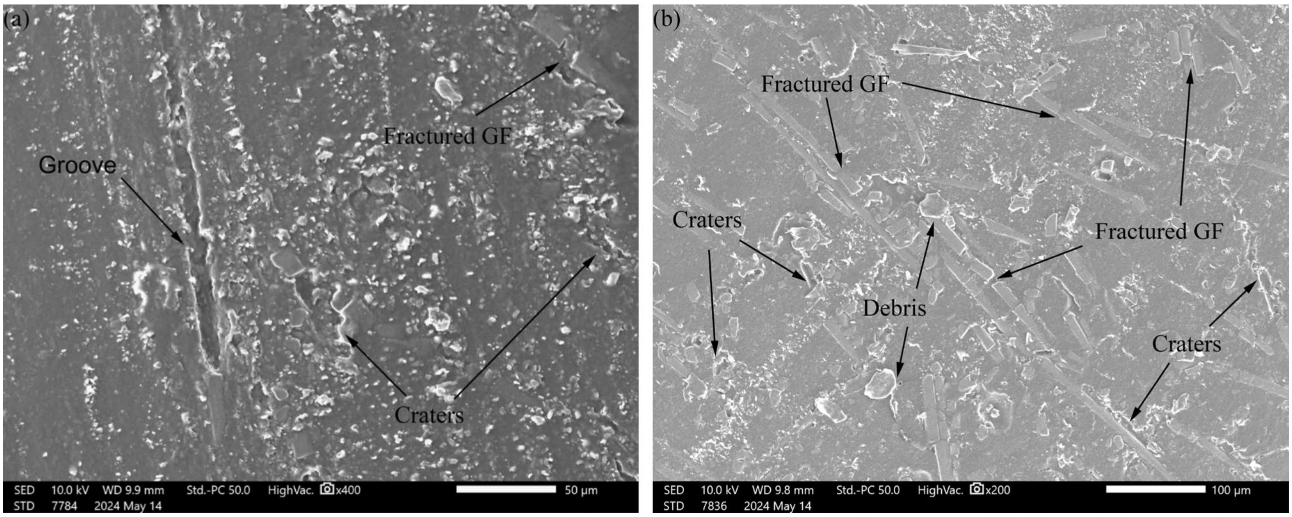

Fig. 7 SEM images of worn surfaces of PA66 composites pins under grease lubricated conditions: **a** PA66 + 13% GF, **b** PA66 + 33% GF (Contact pressure: 60 MPa, sliding speed: 0.5 m/s)

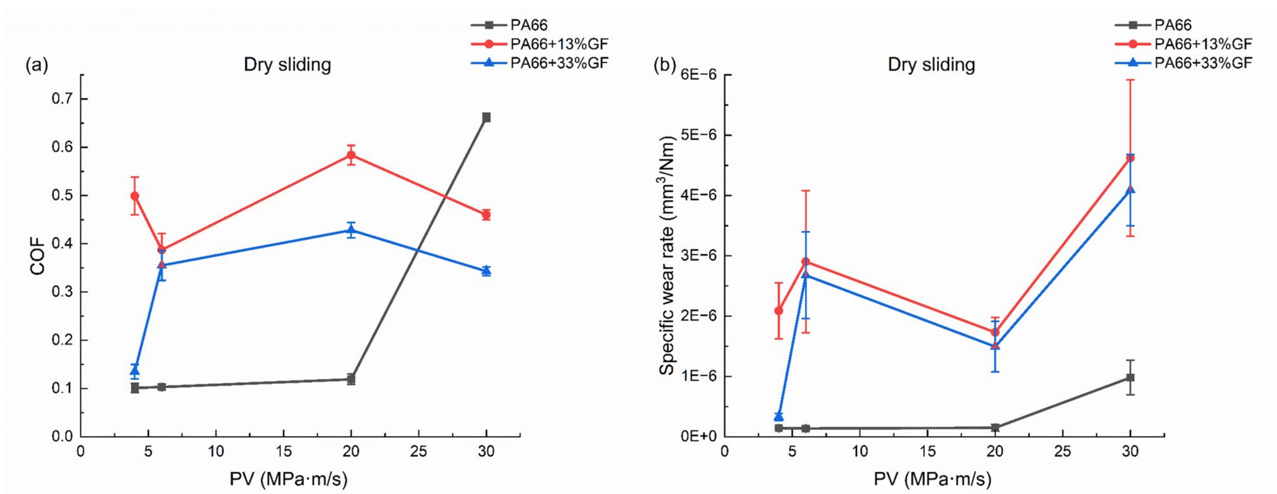

Fig. 8 The coefficient of friction and specific wear rate as a function of PV values under dry sliding conditions: **a** Coefficient of friction; **b** Specific wear rate of pin specimen

3.2 Effect of PV on the Tribological Properties of PA66 and Glass Fiber Reinforced PA66

Figure 8 shows coefficient of friction and the specific wear rate of PA66 composites as a function of PV values under dry sliding conditions. The results show that the coefficient of friction and specific wear rate of PA66 are generally smaller than those of PA66 + 13% GF and PA66 + 33% GF. The coefficient of friction and specific wear rate of PA66 + 33% GF are lower than those of PA66 + 13%GF under all conditions. The specific wear rate and coefficient of friction of PA66 increase with the increase of PV value. The coefficient of friction of PA66 + 13% GF first decreases to a minimum, then increases to a maximum, and finally decreases with the increase of PV value. For PA66 + 33%GF, the coefficient of friction shows an overall increasing trend with increasing PV, but a decrease occurs at 20–30 MPa·m/s. For PA66 + 13%GF and PA66 + 33% GF, the specific wear rate of initially increases, then decreases, and ultimately increases again to the maximum with the increase of PV value. In general, sliding between materials always generates heat, resulting in an increase in the temperature of the sliding surfaces. The generated frictional heat is determined according to the equation $Q = \mu PV$ [29, 37], where μ is the coefficient of friction, P is the applied pressure, and V is the sliding velocity. This means that speed and pressure can

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influence the generation of frictional heat from the sliding process. The smaller coefficient of friction and wear of PA66-PA66 contact combination at smaller PV values can be attributed to insignificant frictional heat accumulation, whereas high frictional heat accumulation at high PV values leads to dramatic increases in friction and wear. Similarly, the gradual accumulation of heat has the same effect on the variation of the coefficient of friction of PA66 + 33%GF with the PV value. The coefficient of friction of PA66 + 13% GF shows different results. It is further found that the results of coefficient of friction for PA66 + 13%GF are the same as those found in references [32, 34, 38, 39], the coefficient of friction decreases with the increase in pressure. This result is in accordance with the model of load and friction coefficient: $\mu = k \cdot F^{n-1}$ [32], where μ is the coefficient of friction, F is the applied load, k and n are associated with the friction system, $k > 0, 2/3 < n < 1$. As the PV value increases, PA66 + 13% GF and PA66 + 33% GF exhibit the highest specific wear rate at a PV value of 30 MPa·m/s. It can be presumed that due to the accumulation of frictional heat at high PV values, the surface of the specimen melts, resulting in severe wear. The specific wear rates of PA66 + 33% GF at PV values of 6 MPa·m/s and 20 MPa·m/s are both greater than the specific wear rate at PV of 4 MPa·m/s, but the specific wear rate at PV of 20 MPa·m/s is lower than the specific wear rate at PV of 6 MPa·m/s. A smaller specific wear rate is also observed at a PV value of 20 MPa·m/s for PA66 + 13% GF. These PV values correspond to different pressures and velocities. It can be assumed that the pressure and speed affect the specific wear rate of PA66 + 13% GF and PA66 + 33% GF [18–21]. And comparing the specific wear rate of glass fiber reinforced PA66 composites at the same speed or the same pressure, it was found that higher pressure and higher speed both increase the specific wear rate. The lower specific wear rate of glass fiber reinforced PA66 at 20 MPa·m/s can be attributed to the lower pressure leading to lower wear.

Figure 9 shows the coefficient of friction and specific wear rate as a function of PV values under grease lubricated conditions. Unlike the results of dry sliding conditions, severe wear does not occur at higher PV values. The coefficient of friction and specific wear rate of the specimens are maximized at the lowest PV under grease lubricated conditions. For PA66, the coefficient of friction and specific wear rate decrease with the increase of PV and are lowest at PV of 20 MPa·m/s, and then slightly increase at PV of 30 MPa·m/s. The coefficient of friction of PA66 + 33% GF first decreases with the increase of PV, with a minimum at PV of 6 MPa·m/s, and then increases with the increase of PV. The trend of specific wear rate of PA66 + 33% GF with the increase of PV is the same as that of the results for PA66 under grease lubricated conditions. For PA66 + 13% GF, the coefficient of friction and specific wear rate initially decrease, then increase, and finally exhibit a minimum of coefficient of friction and specific wear rate at PV of 30 MPa·m/s. It is hypothesized that the frictional heat generated during sliding can be dissipated through the grease [12–14]. In general, it is more likely to lead to the formation of a lubricant film at higher velocities, which protects the specimen and reduces the wear [40, 41]. At PV of 4 MPa·m/s, the lower speed results in thinner lubricant film formation. The pressure on the friction pair is mainly carried directly by the asperities of the contact surfaces, leading to higher wear and coefficient of friction.

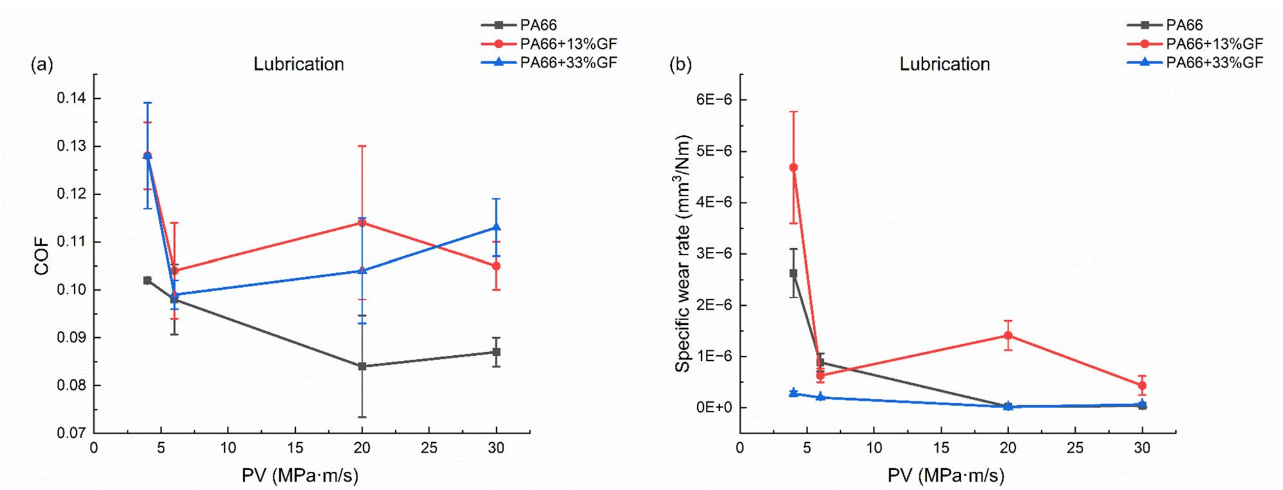

Fig. 9 The coefficient of friction and specific wear rate as a function of PV values under grease lubricated conditions: **a** Coefficient of friction; **b** Specific wear rate of pin specimen

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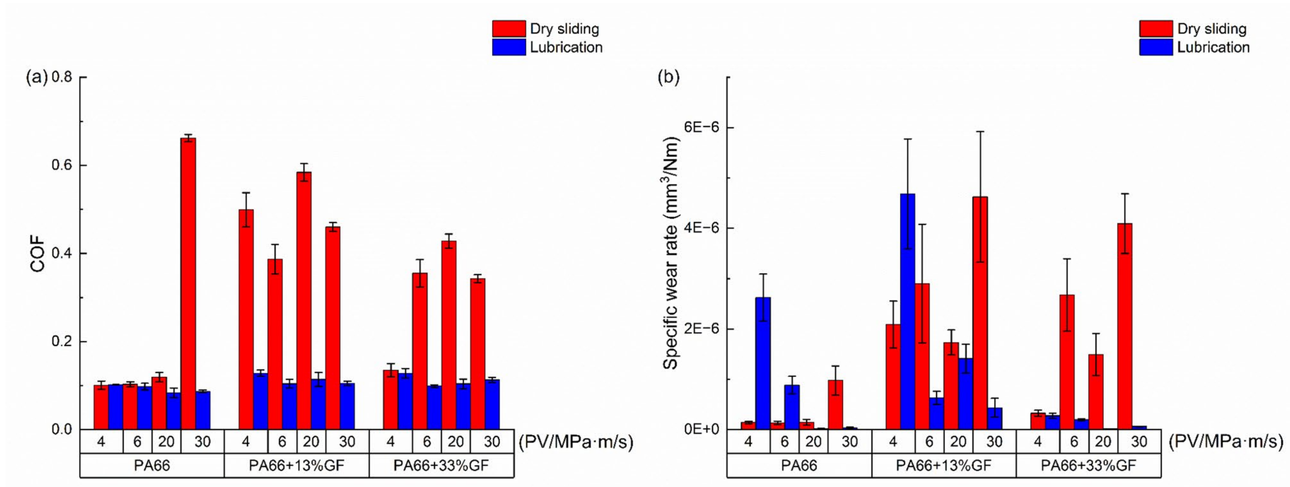

Fig. 10 The tribological properties of the specimens under both dry sliding and lubricated conditions: **a** Coefficient of friction; **b** Specific wear rate of pin specimen

Table 4 The results of the nano-indentation test

Material	Hardness, MPa	Hardness (After lubrication), MPa
PA66	112.8 (111.6 ~ 113.6)	109.1 (104.3 ~ 111.4)
PA66 + 13% GF	98.4 (96.3 ~ 108.1)	94.8 (89.8 ~ 96.1)
PA66 + 33% GF	92.5 (83.7 ~ 94.3)	100.5 (97.1 ~ 108.4)

3.3 Effect of Lubrication on the Tribological Properties of PA66 and Glass *Fiber* Reinforced PA66

Figure 10 shows the coefficient of friction and specific wear rate of the PA66 composites under dry sliding and grease lubricated conditions. The results show that the coefficients of friction of the specimens in the grease conditions are lower than those in the dry sliding conditions in most cases. It is evident that the grease has an obvious effect of reducing the coefficient of friction [12–14, 28]. Furthermore, the grease was found to reduce the wear for the specimens under most experimental conditions. However, at the two PV values (4 MPa·m/s and 6 MPa·m/s) at low speed, the specific wear rates of PA66 and PA66 + 13% GF under grease lubricated conditions are relatively higher than those in the sliding conditions. According to the previous sections, this may be attributed to the fact that the lubricant film thickness is small at low-speed conditions, resulting in the load being carried mainly by the asperities on the specimen surface [36, 40, 41]. Moreover, the presence of grease leads to a reduction in the strength of the sample surface [29]. The results of the three materials after 20 h of grease immersion are shown in Table 4. The hardness of PA66 and PA66 + 13% GF decreased after grease immersion, whereas the hardness of PA66 + 33% GF increased after grease immersion. The weakening effect of grease on PA66 and PA66 + 13% GF results in higher wear under grease lubricated conditions compared to dry sliding conditions. The increased hardness of PA66 + 33% GF due to the presence of grease is likely due to certain interactions between the grease and the PA66 + 33% GF, which change its molecular structure, enhancing its mechanical properties and accounting for its better tribological performance under grease lubrication conditions. The effect of grease on the molecular structure of the specimens will be investigated in subsequent studies. In the high-speed case, the formation of lubricant film is facilitated, with the load is mainly carried by the lubricant film. Consequently, only a fraction of the asperities on the contact surface are involved in the contact, leading to a reduction in the friction and wear of the PA66 composites.

3.4 Damage Mechanism

Figure 11 shows the SEM images of the worn surfaces of PA66 at all PV values under dry sliding conditions. The SEM images are consistent with the results in Fig. 8 (b), indicating that the wear increases with the increase of the PV value. A small number of debris can be observed in Fig. 11a, signifying the domination of the slight adhesive wear mechanism [25]. In Fig. 11b, a large number of debris and craters resulting from material transfer can be observed, marking severe adhesive wear. Slight scratches showing sliding direction are also visible. In Fig. 11c, traces of debris and material transfer affected by the adhesive wear mechanism can be observed. However, clear scratches marking the transition to abrasive wear are also observed. As described in the previous sections, frictional heat accumulates with increasing PV. The increased temperature of the pin surface

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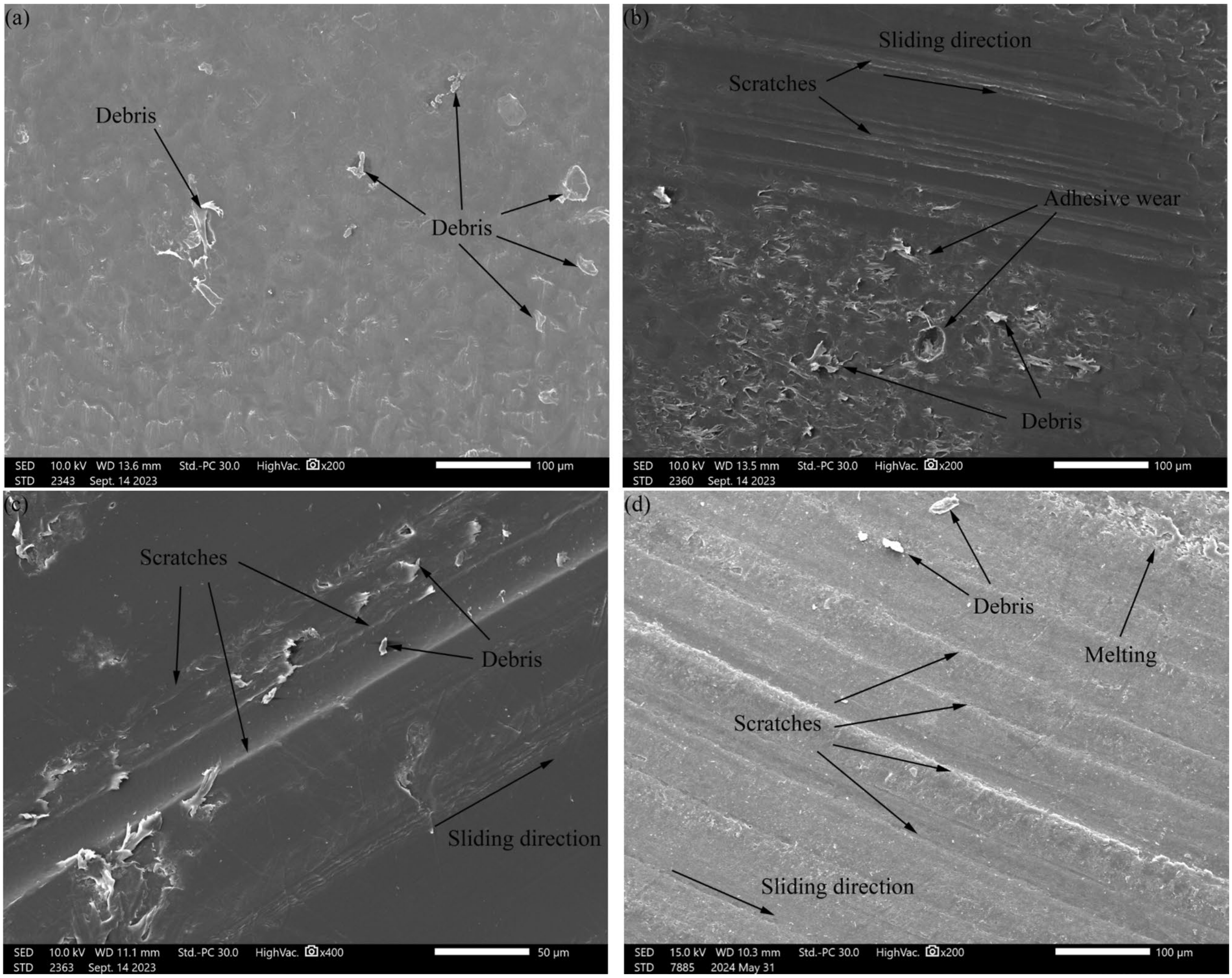

Fig. 11 SEM images of the worn surfaces of PA66 pins under dry sliding conditions: **a** contact pressure: 40 MPa, sliding speed: 0.1 m/s; **b** contact pressure: 60 MPa, sliding speed: 0.1 m/s; **c** contact pressure: 40 MPa, sliding speed: 0.5 m/s; **d** contact pressure: 60 MPa, sliding speed: 0.5 m/s

leads to a decrease in shear strength and material transfer occurs during sliding [29, 37]. The moving disk dissipates heat relatively well, resulting in a lower strength drop and higher hardness relative to the pin, which may lead to abrasive wear on the softer pin surface [28, 42]. In Fig. 11d, a small number of debris, melting due to the accumulation of frictional heat, and a large number of scratches can be observed, indicating the predominance of the abrasive wear mechanism.

Figure 12 shows the SEM images of the worn surfaces of PA66 at all PV values under grease lubricated conditions. In Fig. 12a, a large number of scratches can be observed, marking a predominance of abrasive wear mechanism. This is consistent with the previous results of the specific wear rate in Fig. 9b, where wear is more severe at low PV value. In Fig. 12b, a small number of debris and smearing can be observed, marking the predominance of the adhesive wear mechanism. A small number of debris and slight traces of smearing can be observed in Fig. 12c, marking slight adhesive wear. In Fig. 12d, traces of material transfer, debris and smearing traces can be observed, signifying the domination of the adhesive wear mechanism.

Figure 13 shows the SEM images of the worn surfaces of PA66 + 13% GF at all PV values under dry sliding conditions. The comparison of the worn surface for different PV values is generally in agreement with the results of specific wear rate in Fig. 8b. The specific wear rate is relatively lower for the two PV values in the low-pressure case compared to the two PV values in the high-pressure case. Figure 13a and Fig. 13c shows relatively less wear on the worn surface compared to Fig. 13b and Fig. 13d. In addition, material transfer can be seen in Fig. 13a and Fig. 13c, presumably initially dominated by adhesive wear. During the sliding process, the frictional heat gradually

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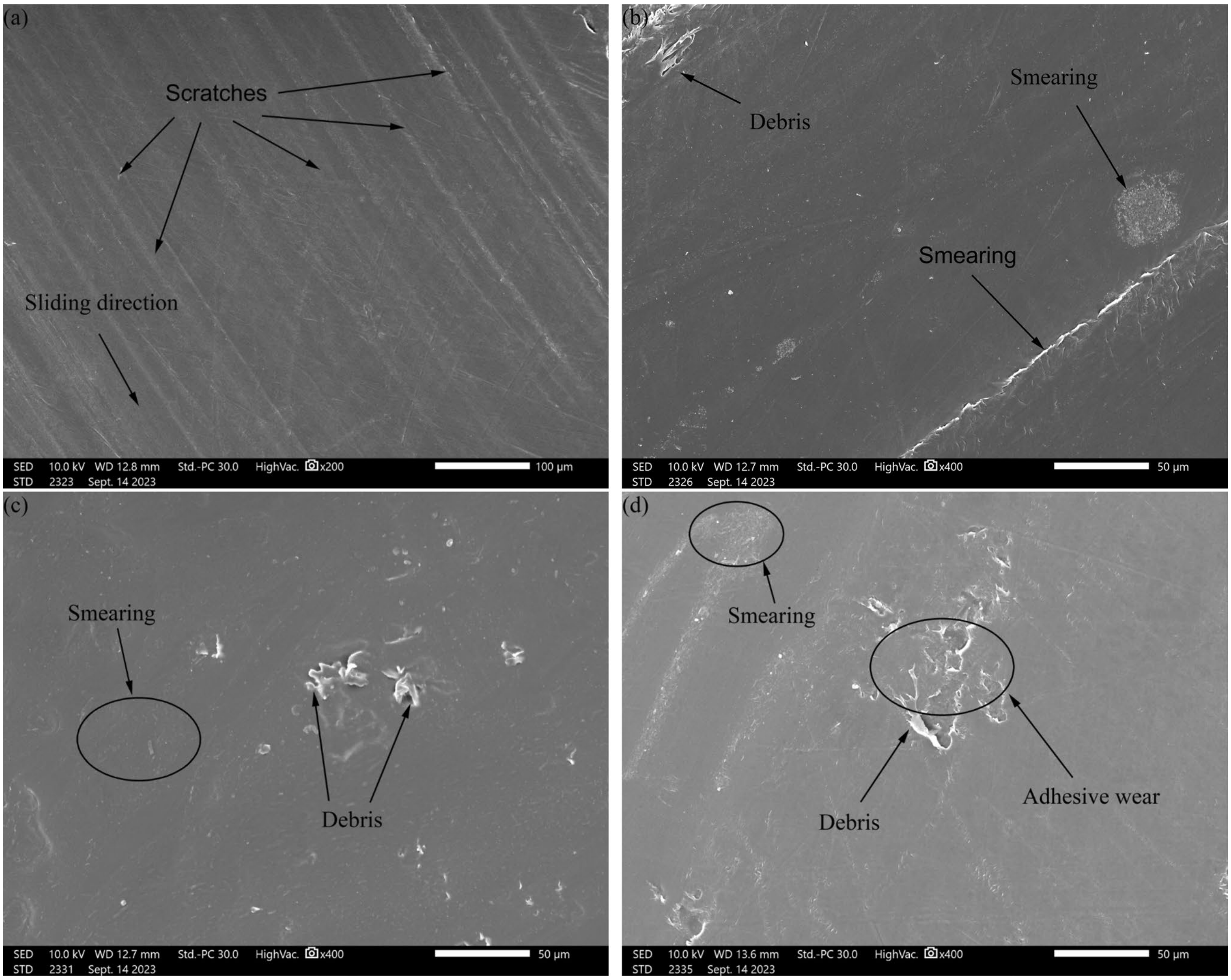

Fig. 12 SEM images of the worn surfaces of PA66 pins under grease lubricated conditions: **a** contact pressure: 40 MPa, sliding speed: 0.1 m/s; **b** contact pressure: 60MPa, sliding speed: 0.1 m/s; **c** contact pressure: 40 MPa, sliding speed: 0.5 m/s; **d** contact pressure: 60 MPa, sliding speed: 0.5 m/s

accumulates and the polymer surface temperature rises, causing it to soften and melt. Some softened and melted materials transfers to counter surface under the action of the shear force. In Fig. 13a-d, fractured glass fibers can be seen, and even the craters and voids produced after the glass fibers is detached from polymer matrix can be observed. As mentioned in the previous sections, the fractured glass fibers act as abrasive particles during the sliding process [17], causing an aggressive effect on the counter surface [12], resulting in a high wear. Moreover, scratches and grooves can be observed in Fig. 13a-c, indicating a change from the initial adhesive wear to the abrasive wear mechanism. The results in Fig. 13b and Fig. 13d show that relatively more glass fibers are broken on the worn surface at the contact pressure of 60 MPa. This is due to the fact that the glass fibers carry most of the load during sliding [10, 17], leading to more severe destruction of the glass fibers, and fragmented glass fibers are visible on the worn surface.

Figure 14 shows the SEM images of the worn surfaces of PA66 + 13%GF at all PV values under grease lubricated conditions. In Fig. 14a, grooves and scratches indicating the sliding direction can be clearly observed, which can be hypothesized to be dominated by the abrasive wear mechanism. Additionally, a small portion of debris associated with adhesive wear is also observed. As mentioned in the previous section, the surface of specimen has been softened by the grease [29], and the wrinkles suggestive of a softening phenomenon can be observed on the surface. The large number of scratches produced by abrasive wear is consistent with the results of the high specific wear rate under grease lubricated conditions at low PV values. In Fig. 14b, a large

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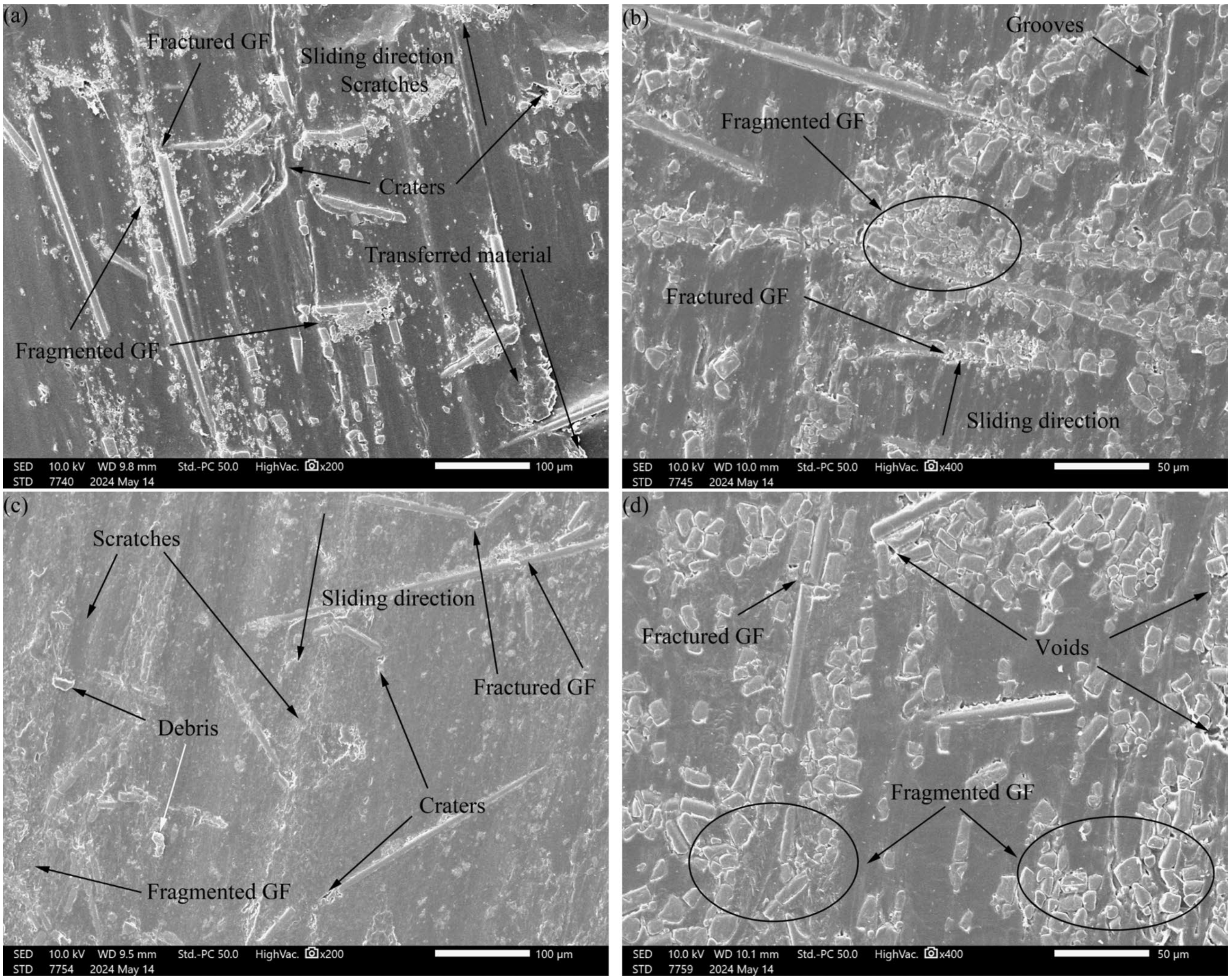

Fig. 13 SEM images of the worn surfaces of PA66 + 13% GF pins under dry sliding conditions: **a** contact pressure: 40 MPa, sliding speed: 0.1 m/s; **b** contact pressure: 60 MPa, sliding speed: 0.1 m/s; **c** contact pressure: 40 MPa, sliding speed: 0.5 m/s; **d** contact pressure: 60 MPa, sliding speed: 0.5 m/s

number of curled debris can be seen that appear to be about to be transferred, suggesting domination by the adhesive wear mechanism. Additionally, glass fibers broken at high contact pressures can also be faintly observed. Most of the polymer matrix debris and glass fibers debris can be seen in Fig. 14c, suggesting domination by the adhesive wear mechanism. However, the observation of craters formed by the detachment of glass fibers suggests a plausible transition toward more severe abrasive wear [17], which is supported by the correlation with previous findings regarding the specific wear rate in Fig. 9b. In Fig. 14d, the fracture of glass fibers under high pressure is more clearly visible than in Fig. 14b, and some craters formed by the pulled-out glass fibers can also be observed.

The worn surface images of PA66 + 33%GF at all PV values under dry sliding conditions are shown in Fig. 15. A large number of debris can be observed in Fig. 15a, suggesting that it is mainly dominated by the adhesive wear mechanism. Fractured glass fibers inside the substrate and craters where the material has detached are also observed. In Fig. 15b, glass fibers fragmented and fractured due to high contact pressure have occurred. Grooves are observed between the polymer matrix, indicating domination by abrasive wear mechanisms, but a small fraction of debris is also observed. In Fig. 15c, fractured and fragmented glass fibers, debris, and craters formed by the pulled-out glass fibers can be observed, suggesting that adhesive wear occurs with material transfer. While the transferred glass fibers will act as abrasive particles on the sliding surface, damaging the

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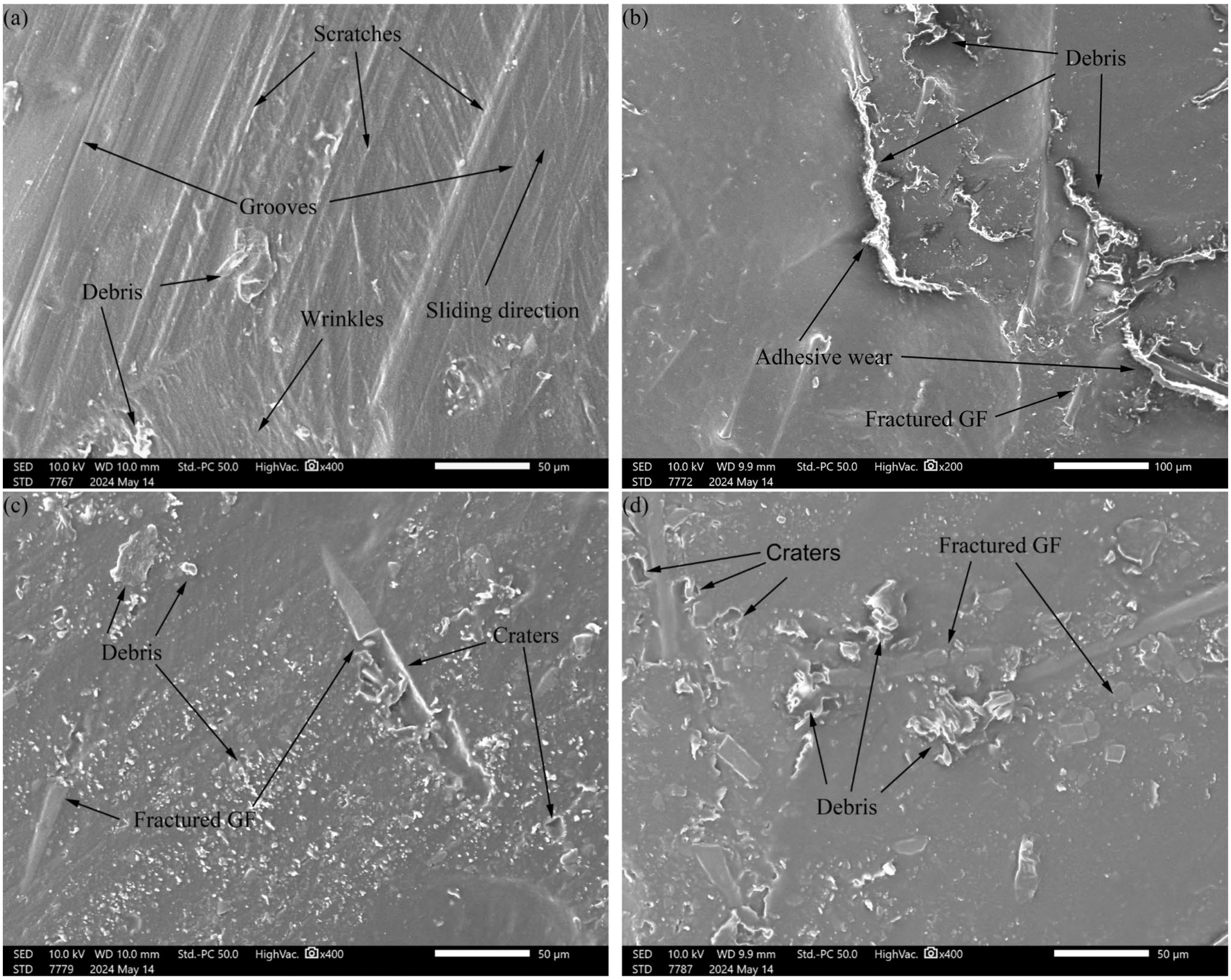

Fig. 14 SEM images of the worn surfaces of PA66 + 13% GF pins under lubricated conditions: **a** contact pressure: 40 MPa, sliding speed: 0.1 m/s; **b** contact pressure: 60 MPa, sliding speed: 0.1 m/s; **c** contact pressure: 40 MPa, sliding speed: 0.5 m/s; **d** contact pressure: 60 MPa, sliding speed: 0.5 m/s

contact surface and shifting toward the abrasive wear mechanism [32, 33]. In Fig. 15d, a large number of fractured glass fibers due to high pressure, and craters formed by peeled-off glass fibers, compressed flakes of glass fibers or polymer matrix, and debris are visible. The glass fibers and polymer matrix are more severely damaged compared to the wear results in the low speed and high load case, which is consistent with the aforementioned high specific wear rate results at high PV values in Fig. 8b.

Figure 16 shows SEM images of the worn surfaces of PA66 + 33% GF at all PV values under grease lubricated conditions. A large number of debris, suggestive of adhesive wear mechanism, can be seen in Fig. 16a, as well as some craters formed due to material transfer. In Fig. 16b, craters formed by broken glass fibers can be observed, along with transferred material debris and smearing, which can be inferred to be mainly dominated by the adhesive wear mechanism. Slight adhesive wear can be seen in Fig. 16c: a small portion of debris, smearing traces, and craters formed by material transfer. Figure 16d also shows evidence of adhesive wear, particularly where glass fibers carried high contact pressure. Craters formed by the peeled-off glass fibers and a portion of the debris can be surmised to be mainly dominated by adhesive wear.

During dry sliding, the worn surface of the PA66 composites is dominated by abrasive wear in most cases. As heat accumulates during sliding, the surface temperature of the

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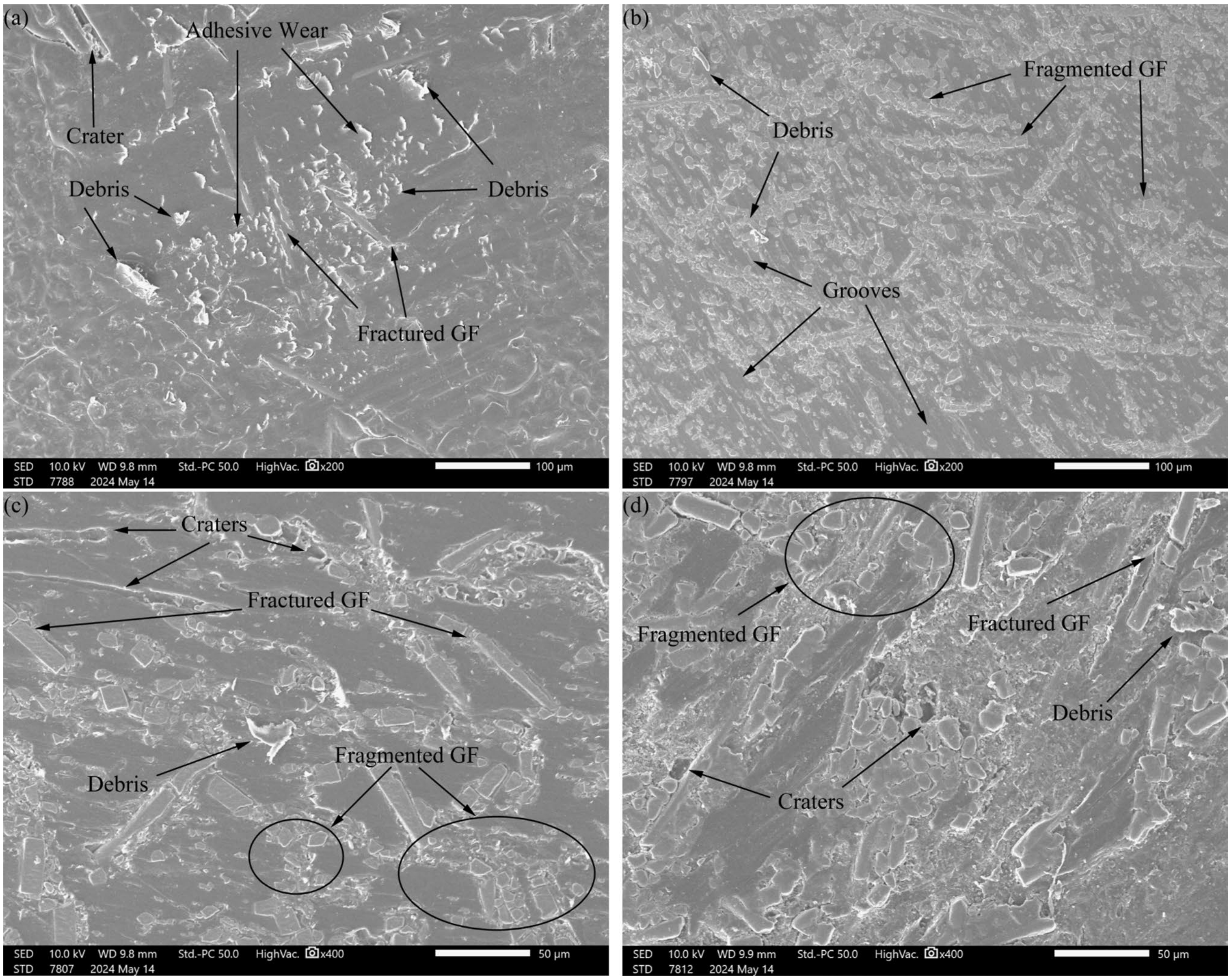

Fig. 15 SEM images of the worn surfaces of PA66 + 33% GF pins under dry sliding conditions: **a** contact pressure: 40 MPa, sliding speed: 0.1 m/s; **b** contact pressure: 60 MPa, sliding speed: 0.1 m/s; **c** contact pressure: 40 MPa, sliding speed: 0.5 m/s; **d** contact pressure: 60 MPa, sliding speed: 0.5 m/s

polymer pin increases, resulting in a reduction in strength and softening of the surface layer. The moving disc dissipates heat more effectively, resulting in a higher hardness, leading to abrasive wear on the softer pin surface. In addition, the adhesion between the glass fibers and the polymer matrix is reduced, which can lead to glass fiber peeling and produce an aggressive effect on the surface, resulting in severe abrasive wear. The peeling of glass fibers may lead to a transition from adhesive wear to abrasive wear in glass fiber reinforced PA66. In contrast, under grease lubricated conditions, worn surface of the composites is predominantly subject to adhesive wear in most cases. This may be due to the fact that grease reduces friction and wear, and also improves heat dissipation. Grease can form a grease film that protects the specimen surface and prevents the glass fibers from detaching from the polymer matrix, thereby improving the tribological properties of PA66 composites.

4 Conclusion

In this study, the sliding part and Hertzian contact of the gear meshing was simulated using a pin-on-disk tribometer. The effects of glass fiber content, PV values (contact pressure and sliding speed), and grease lubrication on tribological properties of PA66 composites in self-mated contact were studied and the following conclusions were drawn:

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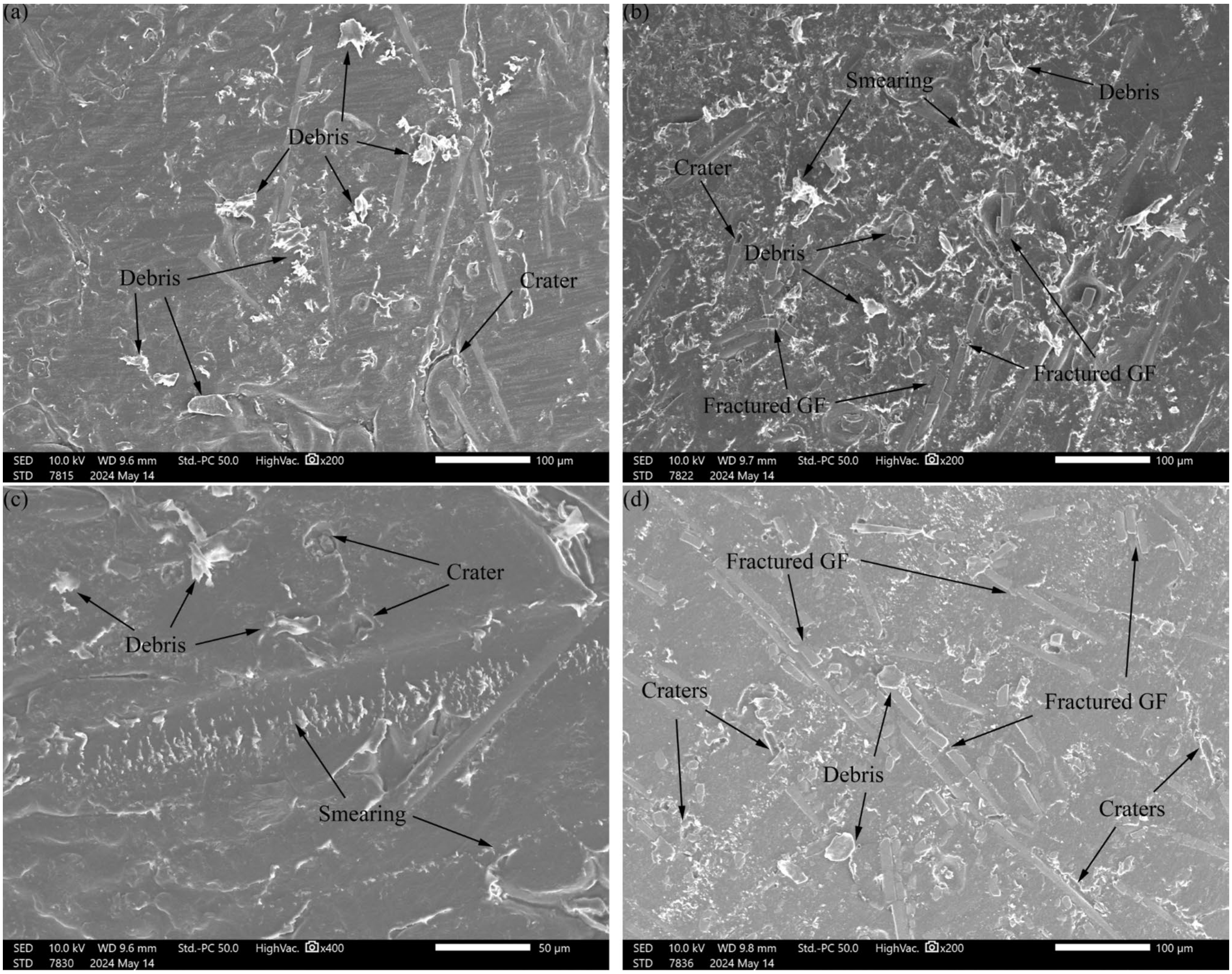

Fig. 16 SEM images of the worn surfaces of PA66 + 33% GF pins under lubricated conditions: **a** contact pressure: 40 MPa, sliding speed: 0.1 m/s; **b** contact pressure: 60 MPa, sliding speed: 0.1 m/s; **c** contact pressure: 40 MPa, sliding speed: 0.5 m/s; **d** contact pressure: 60 MPa, sliding speed: 0.5 m/s

(1) Under dry sliding conditions, the glass fiber composites exhibit higher specific wear rates and coefficients of friction. This is attributed to the peeling of glass fibers during sliding, which leads to increased wear on the contact surfaces.
(2) Under grease lubricated conditions, PA66 + 33% GF exhibits optimal tribological properties due to the fact that grease lubrication reduces the adhesion of the contact surfaces, thus preventing the peeling of glass fibers.
(3) Contact pressure and sliding speed have contrary effect on the wear of the composites. Under dry sliding conditions, the wear of the composites increases with increasing contact pressure and decreases with increasing sliding speed. Contact pressure and sliding speed, known as PV values, also affect the frictional heat accumulation in the sliding contact region, which affects the surface strength of the specimen.
(4) Grease can reduce friction and wear under most experimental conditions. On the one hand, grease reduces the frictional heat during sliding. On the other hand, the grease can form a lubricant film and reduce the adhesion of the polymer–polymer self-contact. The higher wear of PA66 and PA66 + 13% GF at low PV values under grease lubricated conditions may be due to the reduction of mechanical strength.

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Author Contribution X.L., U.O. and L.M. were responsible for conceptualization and methodology design. Z.Q. and X.L. were responsible for the investigation. L.X. and Z.Q. analyzed the experimental data and wrote original draft. All authors reviewed and edited the paper.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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